

1. Administrative & Study Identification

Full / Official Study Title: Real-World Study of Ceftazidime-Avibactam to Characterize the Usage in Clinical Practice
Short Title/Acronym: Real-World Study of Ceftazidime Avibactam in China
Posting ID: 255529375861257558
Protocol Number: C3591037
NCT Number: NCT05487586
Sponsor / Company: Pfizer
Trial Status: TERMINATED (Study was terminated early)
Phase of Development: Not Applicable
Investigational Product (Preferred UMLS Name): Ceftazidime-avibactam (CAZ-AVI)
Drug Name: Ceftazidime
Trade Names: Tazidime, Tazicef, Fortaz, Ceptaz, Avycaz
Medical Monitor: Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe the real-world usage, effectiveness, and antimicrobial features of ceftazidime-avibactam (CAZ-AVI) in clinical practice in China.
Secondary Objectives: To evaluate clinical success rates across specific timepoints and evaluate dosing, administration frequency, and duration of exposure.

2.2 Background & Rationale Infections due to multidrug- and extensively drug-resistant Gram-negative bacteria are difficult to treat and associated with high mortality and disease burden. Ceftazidime-avibactam is an approved combination therapy for such infections. This observational study aims to expand existing real-world evidence by capturing the usage patterns and clinical success rates of CAZ-AVI in routine clinical practice across Chinese hospitals. Patients are followed from CAZ-AVI initiation until death, withdrawal from the study, or 60 days after discharge from hospitalization, whichever comes first.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Prospective, Multicenter)
Control Type: Single-arm (No control; purely observational of standard practice)
Blinding: Open-label
Randomization: N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 220 patients (The study initially planned for approximately 450 in-patients, but was terminated early)
Number of Sites: Approximately 20 clinical research centers in China
Estimated Study Start: 20-Oct-2022
Estimated Primary Completion: 31-Oct-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Initiate ≥ 1 dose of ceftazidime-avibactam during hospitalization. 2. Aged ≥ 18 years old at the time of the informed consent signature. 3. Provide signed informed consent.

4.2 Exclusion Criteria 1. Are enrolled in any clinical trial, including enrollment in non interventional studies. 2. Pregnant women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ceftazidime-avibactam administered per standard clinical practice.

ATC Code: J01DD02

Route of Administration: Intravenous (IV).

Duration of Treatment: As determined by standard clinical practice until the End of Treatment (EOT) or discharge.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant procedures and therapies for related infections and co-morbidities as prescribed by the HCP.

Prohibited: Concurrent enrollment in clinical trials, including non-interventional studies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Baseline	Day 0	Informed Consent, Medical History, Baseline Pathogen/Susceptibility Testing, First Dose CAZ-AVI
Interim	Days 7, 14, 21, 30, 60	In-hospital monitoring: Clinical outcome assessment, Microbiological testing, Concomitant procedures tracking
End of Treatment	EOT	Final in-hospital efficacy assessment, Resistance profiling
Follow-Up	Day 30 & 60 Post-Discharge	Survival status, Final safety and clinical outcome evaluation

(Note: Endpoint events will be evaluated at 7 days, 14 days, 21 days, 30 days, 60 days, and end of treatment (EOT) after CAZ-AVI initiation if patients are not discharged prior to the next upcoming timepoint; and 30 days, 60 days after discharge.)

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Clinical Success Rate at Day 7:** The clinical success rate was defined as the number of participants with clinical outcome as "success" in a specific visit / number of participants with clinical outcome assessed in a specific visit and was reported in terms of percentage of participants. Clinical outcome as success was defined as resolution of all signs and symptoms of infection such that no further antimicrobial therapy was necessary. 95% CI was based on Clopper-Pearson method.
- 2. Clinical Success Rate at Day 14:** The clinical success rate was defined as the number of participants with clinical outcome as "success" in a specific visit / number of participants with clinical outcome assessed in a specific visit and was reported in terms of percentage of participants. Clinical outcome as success was defined as resolution of all signs and symptoms of infection such that no further antimicrobial therapy was necessary. 95% CI was based on Clopper-Pearson method.
- 3. Clinical Success Rate at Day 21:** The clinical success rate was defined as the number of participants with clinical outcome as "success" in a specific visit / number of participants with clinical outcome assessed in a specific visit and was reported in terms of percentage of participants. Clinical outcome as success was defined as resolution of all signs and symptoms of infection such that no further antimicrobial therapy was necessary. 95% CI was based on Clopper-Pearson method.

7.2 Secondary Outcomes

- 1. Number of Participants According to Dose and Frequency of Ceftazidime-Avibactam:** Full Analysis Set
- 2. Number of Participants According to Dose and Frequency of Ceftazidime-Avibactam:** Clinically Evaluable Analysis Set
- 3. Duration of Exposure to Ceftazidime-Avibactam:** Full Analysis Set

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extracted from routine medical records during the index hospitalization and up to 60 days post-discharge.

Serious Adverse Events (SAEs): Reported as per standard observational study and pharmacovigilance guidelines for approved therapies.

Data Monitoring Committee (DMC): Governed by Pfizer's internal real-world data monitoring and ethics committee guidelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Clinically Evaluable (CE) Analysis Set and Full Analysis Set (FAS).

Sample Size Justification: The targeted enrollment of 220

participants is intended to descriptively characterize the real-world usage and estimate clinical success rates with adequate precision in the Chinese population.

Statistical Methods: Summarized descriptively; 95% Confidence Intervals (CI) for clinical success rates calculated using the Clopper-Pearson method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Pharmacoepidemiology Practices (GPP), the Declaration of Helsinki, and local regulatory requirements.

Monitoring Plan: Centralized monitoring of eCRF data extracted from patients' electronic medical records.

Audit Trail: Source data verification against hospital records, eCRF locking, and rigorous data clarification procedures.

11. Study Identifiers & Governance

Primary ID: C3591037

Posting ID: 255529375861257558

Registry Number: NCT05487586

Sponsor/Lead Organization: Pfizer

Study Title: Real-World Study of Ceftazidime Avibactam in China

Official Regulatory Title: Real-World Study of Ceftazidime-Avibactam to Characterize the Usage in Clinical Practice

Study Status: TERMINATED (Study was terminated early)

12. Clinical Framework (Infectious Disease Specific)

Disease Areas / Conditions: Hospital Acquired Pneumonia, Ventilator Acquired Pneumonia, Complicated Intra Abdominal Infections

Targeted Disease Name: Pneumonia (Preferred Name: Pneumonia)

Semantic Type: Disease or Syndrome

Disease Definition: Infection of the lung often accompanied by inflammation.

Terminology Codes:

MeSH Code: D011014

HPO Code: HP:0002090

SNOMED CT ID:* 233604007

Diagnosis Threshold: Hospitalized patient with confirmed/suspected infection requiring IV antibiotics

Medical Methodology: Standard of Care Antimicrobial Therapy

Optimization Target: Resolution of infection (Clinical Success)

13. Study Procedures & Methodology

Management Pathway: Standard clinical treatment pathway for severe nosocomial and intra-abdominal infections

Education Protocol (HCP): Real-world data collection, protocol adherence, and eCRF training

Education Protocol (Patient): Informed consent process detailing observational data usage

Quality Audit Mechanism: eCRF verification matching standard hospital records and microbiology lab reports

14. Observation & Follow-Up Schedule

Total Duration: From CAZ-AVI initiation up to 60 days post-discharge (or withdrawal/death, whichever comes first)

Onsite Visit Cadence: Evaluations mapped to index hospitalization at Days 7, 14, 21, 30, 60, and EOT

Remote Monitoring Frequency: 30 and 60 days post-discharge (usually via telephone or outpatient medical record review)

Checklist Review Interval: End of Treatment (EOT) and Hospital Discharge

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				